

5th International Workshop on the Twin Peaks of Requirements and Architecture (TwinPeaks 2015)

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Abstract—The relationships and interdependencies between software requirements and the architectures of software-intensive systems are described in the Twin Peaks model. The fundamental idea of the Twin Peaks model is that Requirements Engineering and Software Architecture should not be treated in isolation. Instead, we need to progressively discover and specify requirements while concurrently exploring alternative architectural solutions. However, bridging the gap between Requirements Engineering and Software Architecture has mainly been discussed independently in the respective communities. Therefore, this ICSE workshop aims at bringing together researchers, practitioners and educators from the Requirements Engineering and Software Architecture fields to jointly explore the strong interdependencies between requirements and architecture. Based on the results from previous editions of the workshop, this edition focuses on agile software development contexts and on exploring lightweight techniques for integrating requirements and architectural thinking.

Index Terms—Twin peaks, requirements engineering, software architecture, agile.

I. INTRODUCTION

Activities for Requirements Engineering and Software Architecture are often considered and performed separately. However, drawing a line between them is not desirable since requirements and architecture impact each other. Architecting usually starts as soon as an initial understanding of requirements is achieved. Requirements discovery and architectural design therefore need to be interwoven. This synergistic relationship is described in the Twin Peaks Model [1]. The Twin Peaks model is the theoretical foundation for TwinPeaks 2015.

Based on the outcomes of previous editions of the workshop (which were mostly exploratory in nature) and the outcome of TwinPeaks 2014 (which focused on “Development Techniques and Practices for Twin Peaks” and “Twin Peaks in Context”) [2, 3], the main theme for TwinPeaks 2015 is “TwinPeaks goes Agile”. Agile software development has been adopted widely in the software industry [4] since designing and implementing large-scale and complex software systems requires a balance between fast development and delivering customer value.

The agile software development paradigm breaks with the idea of linear development in which either the requirements are specified and frozen without considering the implications on

architectural design, or the architecture is designed without negotiating and resolving emerged conflicts and trade-offs. This iterative and incremental approach towards software development is also reflected in Twin Peaks model. Therefore, TwinPeaks 2015 explores lightweight techniques for integrating requirements and architectural thinking into the agile process, when to engage in upfront requirements analysis and architectural design, and techniques for incremental delivery.

II. WORKSHOP GOALS

The goal of TwinPeaks 2015 is to provide a venue for researchers, practitioners and educators from the areas of Requirements Engineering and Software Architecture, as well as from general software engineering to discuss their experiences, forge new collaborations, and explore innovative solutions that address the challenges of the Twin Peaks model. The workshop offers an opportunity to become familiar with the relationship between Requirements Engineering and Software Architecture in the greater context of software engineering, rather than in an isolated context of either Requirements Engineering or Software Architecture. The outcomes of the workshop are an improved understanding of key issues and challenges at the intersection of Requirements Engineering and Software Architecture, an understanding of the state of the art in research and practice, and a list of directions towards addressing open issues that we have identified in previous editions of the workshop. This will help us further refine a research agenda by adding a list of concrete actions for future work. Furthermore, we aim at strengthen and expanding the community around the topic that has been growing since previous editions of the workshop.

III. WORKSHOP PARTICIPATION AND ACCEPTED PAPERS

The target audience of TwinPeaks 2015 includes researchers who are developing methods, techniques and tools. We also target practitioners who face the challenge of intertwining requirements and architectures. Furthermore, our target audience includes Requirements Engineering and Software Architecture educators who provide academic or professional training. The workshop is open to all ICSE 2015 attendees, but presentations were selected based on reviewing submitted papers. The topics of accepted papers cover a broad

range of issues related to the relationship between requirements and architecture. We included thought-provoking position papers as well as industrial studies. The following papers were accepted for presentation at the workshop (listed in alphabetical order, based on the last name of the first author):

- Preethu Rose Anish, Balaji Balasubramaniam, Jane Cleland-Huang, Roel Wieringa, Maya Daneva and Smita Ghaisas: Identifying Architecturally Significant Functional Requirements
- Erik Hebisich, Matthias Book, Volker Gruhn: Scenario-based Architecting with Architecture Trace Diagrams
- Garm Lucassen, Fabiano Dalpiaz, Jan Martijn van der Werf, Sjaak Brinkkemper: Bridging the Twin Peaks: the Case of the Software Industry
- Paul Ralph, Rahul Mohanani: Is Requirements Engineering Inherently Counterproductive?
- Dimitri van Landuyt, Wouter Joosen: On the Role of Early Architectural Assumptions in Quality Attribute Scenarios: a Qualitative and Quantitative Study

IV. STRUCTURE

We aim at allowing interactive discussions and fostering brainstorming and community building. Therefore, the workshop featured paper presentations which provided the foundations for the participants to investigate open research challenges and practical problems. In addition to the paper presentations based on a review process, the workshop includes a keynote. We are grateful to Jan Bosch from Chalmers University of Technology who agreed to deliver a keynote on “Architecting to Ensure Requirement Relevance”.

V. PROGRAM COMMITTEE

In addition to the two workshop organizers, the program committee consisted of the following:

- Pekka Abrahamsson, University of Bolzano, Italy
- Muhammad Ali Babar, University of Adelaide, Australia

- Jan Bosch, Chalmers University of Technology, Sweden
- Sjaak Brinkkemper, University of Utrecht, Netherlands
- Oliver Creighon, Siemens, Germany
- Maya Daneva, University of Twente, Netherlands
- Christof Ebert, Vector Consulting, Germany
- Neil Ernst, Software Engineering Institute, USA
- Smita Ghaisas, Tata Consulting Services, R&D, India
- Scott Hawker, Rochester Institute of Technology, USA
- Rashina Hoda, University of Auckland, New Zealand
- Anne Koziol, KIT, Germany
- Patrick Maeder, Technische Universitaet Ilmenau, Germany
- Nenad Medvidovic, University of Southern California, USA
- Eltjo Poort, CGI, Netherlands

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